Zcrypt

Check before you act.

تحقّق قبل أن تتصرّف

Zcrypt is a verification tool for the messages and notices people rely on in a crisis. It was developed in Doha in 2026, during a period of regional conflict, to address a simple but serious gap: information that reaches people second-hand cannot easily be trusted, and a forged notice is indistinguishable from a genuine one.

The challenge

In an emergency, critical guidance such as evacuation routes, aid distribution points and school announcements travels through forwarded messages, screenshots and printed notices. Each step strips away the context that shows where it came from.

As recent security incidents and data breaches around the world have shown, current safeguards help but fall short. Verified accounts lose their meaning once a message is forwarded. Forwarding labels show how far a message has travelled, not whether it is authentic. Fact-checking is valuable but typically arrives after people have already acted.

What Zcrypt does

Printed notices

Take a photo. No QR code, no visible mark. If the notice was altered, Zcrypt shows which words changed and what the real notice says.

Messages

A sealed message looks completely normal and keeps its proof when forwarded. Tested this year on WhatsApp and iMessage: sealed messages verify.

Personal documents

Zcrypt Vault keeps a person's certificates on their own phone, recoverable through people they trust, not stored in someone else's database.

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What sets it apart

The words are the proof. Zcrypt verifies the content itself rather than the account, platform or file it arrived in, so a message can be checked wherever it travels.

One method, every channel. The same verification works for WhatsApp and iMessage messages, screenshots and printed notices, in English and Arabic.

It explains, not only flags. When content has been altered, Zcrypt shows exactly what was changed and what the original said, turning a rumour back into accurate information.

Nothing to install. Verification runs in any phone browser, and nothing a person checks is sent to a server.

Built for the quantum era. Zcrypt's core uses post-quantum signatures, designed to remain secure once large-scale quantum computers can break the encryption most systems rely on today.

Prototype status

A working prototype was developed and tested in 2026. In testing, photos of printed notices were verified directly from the printed text, and altered versions were identified with the changed wording highlighted. Sealed messages were verified after being sent over WhatsApp and iMessage.

The web prototype uses standard signatures so that it runs in any browser; Zcrypt's core implementation uses post-quantum signatures based on Falcon, selected by NIST for standardisation.

Next steps are a pilot with a partner organisation, offline verification for low-connectivity settings, and an independent security review.

The value

Most security solutions protect an account, a platform or a file. Zcrypt's value travels with the information itself: it remains verifiable wherever a message goes, it corrects misinformation rather than only flagging it, and it is designed to stay secure for decades, not years.

For organisations. Official guidance remains trustworthy after it leaves official channels, and less personal data needs to be held centrally, reducing the impact of any breach.

For communities. Anyone can check a notice or message independently, in seconds, before acting on it.

For the wider field. A shift in focus from securing systems to securing the information itself, the layer people actually depend on in a crisis.

Try the prototype: scan the code with your phone.

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Zcrypt welcomes conversations with organisations interested in piloting the technology, and with advisors in humanitarian data protection.